DATASCI 185 - Introduction to AI Applications

Danilo Freire

Fall 2026

Course description

Welcome to DATASCI 185! This course offers a non-technical introduction to artificial intelligence and its effects on institutions, work and everyday life. The course is practical: students will learn how modern systems are designed and how to ask the right questions about data, reliability and harms. The course combines short demonstrations (no programming experience required), case studies, and project work. It is suitable for undergraduate students from any faculty who want a grounded understanding of what AI can and cannot do.

Learning objectives

By the end of the course, students will be able to:

- Explain the main ideas behind contemporary AI systems in plain language.
- Identify common failure modes of AI systems and the data issues that cause them.
- Read and assess claims about AI in news articles, product pages and policy documents.
- Design a small, realistic plan for an AI application, including data needs, evaluation, and a basic harm-mitigation strategy.
- Reflect critically on ethical, legal and social questions raised by AI deployment.

Course logistics

- **Lectures:** Mondays and Wednesdays, 4-4:50pm.
- **Location:** Psychology Building, Room 290.
- **Instructor:** Danilo Freire (`danilo.freire@emory.edu`). Office hours by appointment; please email me to arrange a time.
- **Materials:** Lecture notes, tutorials, and assignment templates are on the course GitHub repository. Students must check the repository often for updates.

Prerequisites and software

No formal prerequisites. Familiarity with spreadsheets, basic statistics or introductory programming is useful but not required. We will discuss some simple Python code snippets in class, but no programming experience is needed.

Readings and resources

Readings are short papers, essays and tutorials rather than a single textbook. Links and PDFs will be provided on the course website. Please note that the syllabus may be updated during the semester. For those interested in digging deeper into some of the topics we cover, here are some recommended books and resources:

Books

- Artificial intelligence: a guide for thinking humans by Melanie Mitchell (2020). A clear, non-technical overview of AI concepts and history.
- The master algorithm by Pedro Domingos (2015). A readable introduction to machine learning concepts and applications.
- The coming wave: AI, power, and our future by Mustafa Suleyman and Michael Bhaskar (2025). An award-winning book on the transformative potential of AI and its societal implications.

- The alignment problem: machine learning and human values by Brian Christian (2020). An exploration of what can go wrong when AI systems are deployed in the real world.
- AI & I: An intellectual history of artificial intelligence by Eugene Charniak (2024). A new book by a leading AI researcher, covering the history of the field from the 1950s to the present day.
- Weapons of math destruction: how big data increases inequality and threatens democracy by Cathy O'Neil (2016). A book on the dangers of unregulated mathematical models.
- Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence by Kate Crawford (2021). A critical look at the social and environmental impacts of AI technologies.
- Hands-on large language models: language understanding and generation by Jay Alammar and Marcel van Otterlo (2024). A practical guide to large language models.

Online courses

- AI for everyone by Andrew Ng (Coursera). A non-technical introduction to AI concepts and applications. Free to audit, with a paid certificate option.
- Elements of AI by the University of Helsinki. A free, friendly introduction to AI and some of its methods.
- AI essentials by Google (Coursera). This programme is designed for people who want to gain practical AI skills for the workplace with no experience required.
- Google prompting essentials by Google (Coursera). A short course on how to effectively use and design prompts for large language models.
- Radical ideas in AI ethics by Pragmatic AI Labs (edX). A course that discusses AI through the lens of human rights and digital autonomy.

Other resources

- In machines we trust. A podcast series by MIT Technology Review about the promises and perils of AI. Very accessible and engaging.
- AI for the rest of us. 25-30 minute episodes focused on explaining AI concepts to non-technical listeners.

- The gradient. A publication that features accessible articles on AI research.
- AI Now Institute. Research institute focused on the social implications of AI.
- AI for Good Lab. A Microsoft research group that highlights how AI can change society for the better.
- The AI incident database. A collection of real-world cases where AI systems have caused harm or failed.
- Teachable machine. A web-based tool by Google that allows users to create simple machine learning models without coding.
- The algorithm. A newsletter by MIT Technology Review that covers the latest developments in AI.

Assessments

- **Problem sets (10) — 50%**. Short conceptual and practical tasks. Submit Jupyter notebooks (`.ipynb`), Word documents, or PDFs. Late submissions incur a 10% penalty per day unless authorised in advance. Collaboration for discussion is allowed but answers must be written independently; list collaborators on submission.

- **In-class quizzes (5) — 30%**. Each quiz occupies a full lecture. Quizzes are open-book/open-notes and individual assessments. Discussion during quizzes is not permitted.

- **Final group project — 20%**. Groups of 3–4 produce a report and a short demo video. The project involves conceptual elements and a harms assessment. More details will be provided in class.

Grading scale

Grade	A	A-	B+	B	B-	C	D	F
Range	91–100	86–90	81–85	76–80	71–75	66–70	60–65	<60

Final grades are rounded to the nearest point.

Academic integrity and AI use

Generative AI tools (for example ChatGPT, Claude, GitHub Copilot) may be used as aids for brainstorming or drafting, but all AI use must be declared on submissions. Students must take responsibility for correctness and must attribute any direct text or code produced by AI. The Emory Honour Code applies.

Emory Honour Code

The Emory Undergraduate Academic Honour Code is in effect throughout the semester. The Honour Code applies to any action or inaction that fails to meet the communal expectations of academic integrity. Students should strive to excel in their academic pursuits in a just way with honesty and fairness in mind and avoid all instances of cheating, lying, plagiarizing, or engaging in other acts that violate the Honor Code. Such violations undermine both the individual pursuit of knowledge and the collective trust of the Emory community. Students who violate the Honour Code may be subject to failure of the course, a reportable record, suspension, permanent expulsion, or a combination of these and other sanctions. The Honor Code may be reviewed at: `http://catalog.college.emory.edu/academic/policies-regulations/honor-code.html`.

Accessibility

Students who require accommodations should contact the Department of Accessibility Services early in the semester and provide documentation. Reasonable accommodations will be made for assessments, including quizzes, according to DAS guidance. Please contact me privately to discuss any specific needs.

Course schedule

Please find below the schedule of our lectures. Each lecture includes a brief description, required readings, and suggested readings for further exploration. The lecture slides and additional resources will be posted on the course website and GitHub repository. Please remember that the schedule may change during the course. If you have any questions or need further assistance, please let me know!

Module 0 — Orientation

January 14, 2026:

- Syllabus and course repository: `https://github.com/danilofreire/datasci185`.
- Lecture 01: Course introduction.

Readings:

- Stanford University's Human-Centered Artificial Intelligence (2025). AI Index Annual Report. An excellent report on the state of AI, with many useful charts and references.
- Our World in Data (2025). Artificial intelligence. A concise overview of AI trends and applications, with links to further resources.
- Menlo Ventures (2025). 2025: the state of consumer AI. How AI is being used in consumer products, with examples and market data.
- Morgan Stanley (2025). AI's next leap: 5 trends shaping innovation and ROI. A short text on the current state of AI applications in industry.
- McKinsey & Company (2025). The state of AI: how organizations are rewiring to capture value. A business-focused overview of AI adoption and impact.

January 19, 2026: Martin Luther King Day (No Classes)

January 21, 2026:

- Lecture 02: A brief history of AI and the recent shift.
- **Assignment 01**.

Readings:

- Dick, S. (2019). Artificial intelligence. *Harvard Data Science Review*, *1*(1). A brief history of AI research and applications. Written for a general audience.
- Grzybowski, A., Pawlikowska–Łagód, K., & Lambert, W. C. (2024). A history of artificial intelligence. *Clinics in Dermatology*, *42*(3), 221-229. A "pre-history" overview, focusing on early ideas and milestones.

- Cao, Y., Liu, L., Li, M., Huang, Y., & Li, S. (2023). A comprehensive survey of AI-generated content (AIGC): A history of generative AI from GAN to ChatGPT. *arXiv preprint.* A somewhat technical but comprehensive overview of generative AI methods.
- Halevy, A., Norvig, P., & Pereira, F. (2009). The unreasonable effectiveness of data. *IEEE Intelligent Systems*, *24*(2), 8-12. A classic paper arguing that large datasets are often more important than sophisticated algorithms.
- Pradhan, M. (2023). A non-technical introduction to Transformers. A clear explanation of the transformer architecture that underpins many modern AI systems.
- Georgia Tech's Polo Club of Data Science (2025). Transformer explainer. An interactive visualisation of how transformers work.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., … & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30. In case you want to read the original paper. Not for the faint of heart.

Module 1 — How AI systems are designed

January 26, 2026:

- Lecture 03: Dataset design, labels, and tasks.
- Kahoot Quiz

Readings:

- Athalye, A., Northcutt, C., & Mueller, J. (2024). Dataset creation and curation in *Data-Centric AI*. A chapter from a forthcoming book on data-centric AI, covering the main ideas in dataset design.
- Rich, A. S., & Gureckis, T. M. (2019). Lessons for artificial intelligence from the study of natural stupidity. *Nature Machine Intelligence*, 1(4), 174-180. A title this good is hard to find. The article discusses different sources of bias in AI systems.
- Sapien Labs. (2024). Labeling data for machine learning: best practices and quality control. A practical guide to data labelling, with tips and common pitfalls.

- Liang, W., Tadesse, G. A., Ho, D., Fei-Fei, L., Zaharia, M., Zhang, C., & Zou, J. (2022). Advances, challenges and opportunities in creating data for trustworthy AI. *Nature Machine Intelligence*, 4(8), 669-677. A good discussion on the challenges of creating high-quality datasets.
- Chai, C., & Li, G. (2020). Human-in-the-loop techniques in machine learning. *IEEE Data Eng. Bull.*, 43(3), 37-52. A little more technical, but quite comprehensive.

January 28, 2026:

- Lecture 04: Supervised, unsupervised, and reinforcement learning.
- Kahoot Quiz
- **Assignment 01 due (5%).**
- **Assignment 02**.

Readings:

- Jiang, T., Gradus, J. L., & Rosellini, A. J. (2020). Supervised machine learning: a brief primer. *Behavior Therapy*, *51*(5), 675-687. A non-technical introduction to supervised machine learning.
- Amazon Web Services (2025). What is the difference between supervised and unsupervised learning? A practical overview with examples.
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Chapter 5: Machine learning basics in *Deep Learning*. A technical but clear introduction to the main ideas behind machine learning, by some of the founders of deep learning.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An introduction to statistical learning. A widely used book covering supervised and unsupervised learning methods. Included here for reference, as you'll surely use it in your future career.
- Ghasemi, M., & Ebrahimi, D. (2024). Introduction to reinforcement learning. *arXiv preprint*. Short introduction to the topic. If you would like to read something more advanced, please refer to Sutton & Barto (2015).

February 02, 2026:

- Lecture 05: Metrics, validation and overfitting.

- Kahoot Quiz

Readings:

- Thomas, R. L., & Uminsky, D. (2022). Reliance on metrics is a fundamental challenge for AI. *Patterns*, 3(5).100440. A discussion of the limitations of metrics.
- Akinkugbe, A. (2025). The essential guide to model evaluation metrics for classification. A practical guide to common evaluation metrics.
- Confident AI (2025). LLM evaluation metrics: the ultimate LLM evaluation guide. A simple guide on new ways to measure the quality of LLM outputs.
- Rashka, S. (2025) Understanding the 4 main approaches to LLM evaluation. Good blog post about the different ways to evaluate LLMs.
- Coursera (2025). Precision vs. recall in machine learning: What's the difference?. A short article about those two commonly confused metrics.
- Inie, N., Stray, J., & Derczynski, L. (2023). Summon a demon and bind it: a grounded theory of LLM red teaming in the wild. *arXiv preprint*. A good overview of red teaming.
- Radharapu, B., Robinson, K., Aroyo, L., & Lahoti, P. (2023). Aart: AI-assisted red-teaming with diverse data generation for new LLM-powered applications. *arXiv* preprint*. A method to improve LLM safety with simple adversarial attacks.

Module 2 — Language and perception

February 04, 2026:

- Lecture 06: How machines handle language: tokens, embeddings and context.
- Kahoot Quiz
- **Assignment 02 due (5%).**
- **Assignment 03**.

Readings:

- HuggingFace (2025). Tokenizer summary A really good overview of tokenisation, with examples and videos.

- Neptune.ai (2025). The ultimate guide to word embeddings. A clear and practical introduction to word embeddings.
- Neptune.ai (2025). Tokenization in NLP: types, challenges, examples, tools. Another article by the same group on tokenisation.
- StackOverflow (2023). An intuitive introduction to text embeddings. A practical introduction to embeddings.
- Metzger, S. (2022). A beginner's guide to tokens, vectors, and embeddings in NLP.. Easy to read and with interesting examples.
- Boykis, V. (2022). What are embeddings?. A whole book dedicated to embeddings. For those who are interested in a more in-depth treatment. There's also a GitHub repository with code examples.

February 09, 2026:

- Lecture 07: How machines see and hear.

Readings:

- O'Shea, K., & Nash, R. (2015). An introduction to convolutional neural networks. *arXiv preprint.* A clear and concise tutorial.
- Gupta, A. (2025). How computers see the world: a beginner's guide to CNNs An illustrated introduction to convolutional neural networks. Feel free to skip the coding section if you are not interested.
- AltexSoft (2025). How AI sound, music, and voice generation works. Another practical introduction. It goes a little beyond what we cover in class.
- Raschka, S. (2024). Understanding Multimodal LLMs. More technical but clear.
- Wu, J., Gan, W., Chen, Z., Wan, S., & Yu, P. S. (2023). Multimodal Large Language Models: A Survey. *arXiv preprint.*
- Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2023, July). Robust speech recognition via large-scale weak supervision. In *International conference on machine learning* (pp. 28492-28518). PMLR. The original paper for Whisper.

February 11, 2026:

- Lecture 08: Quiz 01.
- **Assignment 03 due (5%).**
- **Assignment 04**.

February 16, 2026:

- Lecture 09: Prompt engineering, meta-prompting, and AI agents.

Readings:

- Anthropic (2025). Prompt engineering overview. Official guide from Claude's creators.
- OpenAI (2025). Prompt engineering. OpenAI's best practices.
- Google (2025). Gemini for Workspace Prompting Guide. The persona-task-context-format (PTCF) framework.
- Wei, J., et al. (2022). Chain-of-Thought prompting elicits reasoning in large language models. The original CoT paper.
- Prompting Guide. Community-maintained reference with examples.
- Google (2025). Prompting essentials. Free course on Coursera.

Module 3 — Retrieval, generation and pipelines

February 18, 2026:

- Lecture 10: Creativity and hallucination.
- **Assignment 04 due (5%).**
- **Assignment 05**.

Readings:

- Jiang, X., Tian, Y., Hua, F., Xu, C., Wang, Y., & Guo, J. (2024). A survey on large language model hallucination via a creativity perspective. *arXiv preprint.*
- Wikipedia (2025). Hallucination (artificial intelligence). A well-written overview.
- OpenAI (2024). Does ChatGPT tell the truth? Practical advice.

February 23, 2026:

- Lecture 11: Retrieval-augmented workflows and semantic search.

Readings:

- Amazon Web Services (2025). What is retrieval-augmented generation (RAG)? Beginners' guide to RAG.
- Mastering LLM (2024). 11 chunking strategies for RAG — simplified & visualized Nice explanation of chunking strategies.
- Slack (2025). Semantic search explained: how it works, and why it matters. Good explainer.

February 25, 2026:

- Lecture 12: Quiz 02.
- **Assignment 05 due (5%).**
- **Assignment 06**.

March 02, 2026:

- Lecture 13: Building reliable pipelines: monitoring and testing.

Readings:

- Google (2025). Production ML systems: monitoring pipelines. Practical guide to monitoring ML pipelines.
- Allen, J. (2024). Data pipeline monitoring: best practices for full observability. Practical tips.

Module 4 — Data ethics and bias

March 04, 2026:

- Lecture 14: Documentation and dataset governance.
- **Assignment 06 due (5%).**
- **Assignment 07**.

Readings:

- Gebru, T., et al. (2018). Datasheets for datasets. *arXiv preprint.* The original paper proposing datasheets for datasets.
- Mitchell, M., et al. (2019). Model cards for model reporting. *Proceedings of FAccT.* The original paper proposing model cards.
- IBM (2025). What is data lineage? A short guide to data lineage.

March 9 and 11, 2026: Spring Break (No Classes)

March 16, 2026:

- Lecture 15: Types of bias and how they arise.

Readings:

- Barocas, S., Hardt, M., & Narayanan, A. (2019). Fairness and machine learning. Chapter 2: When is automated decision making legitimate?
- Mehrabi, N., Morstatter, F., Saxena, N., Lettich, A., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys*, *54*(6), 1-35.
- Silberg, J., & Manyika, J. (2019). Notes from the AI frontier: tackling bias in AI (and in humans). McKinsey Global Institute.

March 18, 2026:

- Lecture 16: Case studies: biased outcomes in health, hiring and policing.
- **Assignment 07 due (5%).**
- No assignment this week.
- Instructions for the Final Project.

Readings:

- Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, *366*(6464), 447-453.

- Buolamwini, J., & Gebru, T. (2018). Gender shades: intersectional accuracy disparities in commercial gender classification. *Proceedings of Machine Learning Research*, *81*, 1-15.
- Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016). Machine bias. *ProPublica*.

Module 5 — Policy, governance and social impact

March 23, 2026:

- Lecture 17: AI regulation and standards around the world.

Readings:

- The White House. (2025). Winning the race: America's AI action plan. What the US government wants to do about AI.
- European Parliament. (2023). EU AI act. The original text.
- Cole, M. D. (2024). AI regulation and governance on a global scale. *Journal of AI Law and Regulation*, *1*(1), 126-142.

March 25, 2026:

- Lecture 18: Quiz 03.
- **Assignment 08**.

March 30, 2026:

- Lecture 19: Privacy and data protection.

Readings:

- Hoofnagle, C. J., Van Der Sloot, B., & Borgesius, F. Z. (2019). The European Union general data protection regulation: what it is and what it means. *Information & Communications Technology Law*, *28*(1), 65-98.
- Taddeo, M., & Floridi, L. (2018). How AI can be a force for good. *Science*, *361*(6404), 751-752.

April 01, 2026:

- Lecture 20: Labour markets and economic effects.
- **Assignment 08 due (5%).**
- **Assignment 09**.

Readings:

- Acemoglu, D., & Restrepo, P. (2019). Automation and new tasks: how technology displaces and reinstates labor. *Journal of Economic Perspectives*, *33*(2), 3-30.
- World Economic Forum. (2025). The future of jobs report 2025. A comprehensive report on how jobs are expected to change.
- Anthropic (2026). Anthropic economic index. How people use Claude and what are its possible impacts on the economy.
- OpenAI (2025). How people are using ChatGPT. Same thing, but for ChatGPT. Spoiler: most people are not using it for work. The full article is available here.

Module 6 — Applications, limits and projects

April 06, 2026:

- Lecture 21: AI in healthcare and education.

Readings:

- Opel, N., & Breakspear, M. (2026). Transforming mental health research and care through artificial intelligence *Science*, *391*(6782), 249-258. The authors review ways in which AI may reduce care inequities when deployed responsibly. They discuss how AI can be useful across several stages of care, from prevention to treatment and rehabilitation.
- Agentic Health AI. (2025). Awesome AI agents for healthcare. A curated list of research papers, projects, and resources related to the application of AI agents for healthcare, including medical image analysis, EHR manipulation, counseling, drug discovery, patient dialogue, and healthcare administration.

- Habicht, J., Dina, L. M., McFadyen, J., Stylianou, M., Harper, R., Hauser, T. U., & Rollwage, M. (2025). Generative AI–enabled therapy support tool for improved clinical outcomes and patient engagement in group therapy: real-world observational study. *Journal of Medical Internet Research*, *27*, e60435. An experiment on the use of AI in group therapy. It has a positive result.
- Moore, J., Grabb, D., Agnew, W., Klyman, K., Chancellor, S., Ong, D. C., & Haber, N. (2025). Expressing stigma and inappropriate responses prevents LLMs from safely replacing mental health providers. *arXiv preprint arXiv:2504.18412*. The authors question the idea that LLMs should replace mental health providers.
- OpenAI. (2025). Strengthening ChatGPT's responses in sensitive conversations. OpenAI's approach to making ChatGPT more helpful in sensitive conversations.
- Zhai, X., et al. (2021). A review of artificial intelligence (AI) in education from 2010 to 2020. *Complexity*, 2021(1), 8812542.

April 08, 2026:

- Lecture 22: Quiz 04.
- **Assignment 09 due (5%).**
- **Assignment 10**.

April 13, 2026:

- Lecture 23: Misinformation, deepfakes and trust online.

Readings:

- Helmus, T. C. (2022). Artificial intelligence, deepfakes, and disinformation: a primer. *RAND Corporation.*
- Chesney, R., & Citron, D. K. (2019). Deepfakes and the new disinformation war. *Foreign Affairs.*

April 15, 2026:

- Lecture 24: Long-term safety, the alignment problem and the future of AI.
- **Assignment 10 due (5%).**

Readings:

- Amodei, D., et al. (2016). Concrete problems in AI safety. *arXiv preprint.* A highly influential paper on the practical challenges of AI safety.
- Ji, J., Qiu, T., Chen, B., Zhang, B., Lou, H., Wang, K., … & Gao, W. (2023). AI alignment: a comprehensive survey. *arXiv preprint.* A survey of the field of AI alignment.
- Zhi-Xuan, T., Carroll, M., Franklin, M., & Ashton, H. (2025). Beyond preferences in AI alignment. *Philosophical Studies*, 182(7), 1813-1863. A philosophical take on the limits of rational choice in AI alignment. A little more abstract, but very interesting.
- Anthropic. (2025). Constitutional classifiers: defending against universal jailbreaks. A jailbreak defense system using input/output classifiers trained on synthetic data. It reduced jailbreak success rates by a lot… but some people still found ways around it. If you would like to read the whole academic paper, check it out here.
- Russell, S. (2017). 3 principles for creating safer AI. *TED Talk*.

April 20, 2026:

- Lecture 25: Course revision.

April 22, 2026:

- Lecture 26: Quiz 05.

April 27, 2026:

- No lecture.
- Q&A session and final project work time.
- Final Project due (20%).